

Arushi Sagar

Ghaziabad, Uttar Pradesh, India | ✉ arushisagar18@gmail.com | ☎ +91-7488358904

🐙 GitHub | 🔗 LinkedIn

Education

KIET Deemed to be University

Expected Graduation: 2028

B.Tech in Computer Science & Engineering (AI & ML) | CGPA: 8.27

Technical Skills

Programming: Python, C++, SQL

Machine Learning: Supervised Learning, Feature Engineering, Model Evaluation, XGBoost, SHAP

Deep Learning & GenAI: PyTorch, Transformers, NLP, LangChain, LLM Applications, Prompt Engineering, RAG

Data & Visualization: Pandas, NumPy, Plotly, Streamlit, Folium

Backend & APIs: FastAPI, REST APIs, Pydantic

Databases: MySQL, SQLite

Tools & Cloud: Git, GitHub, VS Code, Google Colab, Power BI, AWS

Projects

InfraUrban – Urban Heat Intelligence Platform

Jan 2026 – Mar 2026

GitHub | Live Demo

- Trained an XGBoost heat-risk model across 24 zones in 6 Indian cities using forward time validation, improving RMSE by 61.9% over a lag-1 baseline.
- Built SHAP explainability, warming trend analysis, and deployed results through a Streamlit dashboard with geospatial visualization.

Tech: XGBoost, SHAP, SQLite, Streamlit, Folium, Plotly

CycleSphere – Menstrual Health Analytics Platform

Nov 2025 – Dec 2025

GitHub

- Processed 17,976 cycle records across 2,000 users using pandas and SQL to analyze lifestyle-cycle variability relationships.
- Applied Kruskal-Wallis testing and silhouette-optimized KMeans clustering with PCA for statistical analysis and behavioural segmentation.

Tech: Scikit-learn, SciPy, SQL, Pandas, Streamlit, Plotly

PipelineGuardian – ML Experimentation Auditor

Jul 2026

GitHub | Live Demo

- Built an auditor for ML notebooks and scripts detecting data leakage, reproducibility gaps and validation mismatches.
- Combined AST-based static analysis with targeted LLM reasoning and deployed the system as a CLI, FastAPI backend and web application.

Tech: LangChain, FastAPI, Python AST, Pydantic, Groq, Pytest

Problem Solving & Open Source

- Solved 100+ DSA problems on LeetCode spanning arrays, hashing, trees, graphs, and dynamic programming.
- Merged a feature PR to render-engine (Python static site generator), adding include/exclude glob filtering for static site map entries with test coverage; iterated through 4 rounds of maintainer review.

Certifications

AWS Certified Data Engineer – Associate | AWS Certified Machine Learning – Associate | AWS Certified Cloud Practitioner – Foundational